

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems

COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Short Answer Text(High) Assessment Tool Weightage: 30%

Dr

QUESTIONS		Total Marks: 50	
Q.N o	Question	Bloom's Level	Marks
1	UPS Product Tracking Information System UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries). These transportation events are	BL1 – Remember	5

	characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute. Create an Entity Relationship diagram that captures this information about the UPS system.		
2	Course Management System Consider a university database for the scheduling of classrooms for final exams. This database could be modeled as the single entity set exam, with attributes course-name, section-number, room-number, and time. Alternatively, one or more additional entity sets could be defined, along with relationship sets to replace some of the attributes of the exam entity set, as • course with attributes name, department, and c-number • section with attributes s-number and enrollment, and dependent as a weak entity set on course • room with attributes r-number, capacity, and building Show an E-R diagram illustrating the use of all three additional entity sets listed.	BL2 – Understand	5
3	University Registrar's Office Management System A university registrar's office maintains data about the following entities: - courses, including number, title, credits, syllabus, and prerequisites; - course offerings, including course number, year, semester, section number, instructor(s), timings, and classroom; - students, including student-id, name, and program; - instructors, including identification number, name, department, and title. - Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled. Construct an E-R diagram for the registrar's office.	BL2 – Understand	5

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4	Hospital Management System Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Construct appropriate tables for the above ER Diagram : Patient(SS#, name, insurance) Physician (name, specialization) Test-log(SS#, test-name, date, time) Doctor-patient (physician-name, SS#) Patient-history(SS#, test-name, date)	BL2 – Understand	5
5	Company Project Management System Construct an ER Diagram for Company having following details : Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations. • A department controls a number of PROJECT. Projects have a unique name, number and a single location. • Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee. • Employee's DEPENDENT are tracked for health insurance purposes (dependentname, birthdate, relationship to employee).	BL2 – Understand	5
6	Explain schema and instance with examples. Distinguish between logical schema, physical schema, and view schema.	BL1 – Remember	5
7	Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.	BL2 – Understand	5

8	Explain the Extended ER (EER) model. Describe the concepts of generalization, specialization, and aggregation.	BL2 – Understand	5
9	Design an EER diagram for a Hospital Management System incorporating inheritance and weak entities.	BL2 – Understand	5
10	Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.	BL1 – Remember	5

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Linkages	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Organization	20%	Correctly applies concepts to solve the	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application

Code of Conduct:

I M.Dhruthika (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

		question/problem			
Integration of Literature	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS:

1. UPS Product Tracking Information System (ER Diagram)

Entities

Shipped_Item

- **ItemNo (PK)**
- Weight
- Dimensions
- InsuranceAmount
- Destination
- FinalDeliveryDate

Retail_Center

- **CenterID (PK)**
- Type
- Address

Transportation_Event

- **ScheduleNumber (PK)**
- Type (Flight/Truck)
- DeliveryRoute

Relationships

- **Received_At**
 - Retail_Center (1) — (M) Shipped_Item
 - One retail center receives many shipped items.
 - Each shipped item is received at only one retail center.
- **Transported_By**
 - Shipped_Item (M) — (M) Transportation_Event
 - A shipped item can travel through multiple transportation events.
 - One transportation event carries many shipped items.

ER Diagram (Text)

Retail_Center

CenterID (PK)

Type

Address

|

| Receives (1:M)

|
Shipped_Item

ItemNo (PK)
Weight
Dimensions
InsuranceAmount
Destination
FinalDeliveryDate

|
| Transported_By (M:N)
|

Transportation_Event

ScheduleNumber (PK)
Type
DeliveryRoute

2. Course Management System (ER Diagram)

Entities

Course

- Name (PK)
- Department
- CourseNumber

Section (Weak Entity)

- SectionNumber (Partial Key)
- Enrollment

Room

- RoomNumber (PK)
- Capacity

- Building

Exam

- Time

Relationships

- Course **Has** Section (1:M)
- Section **Scheduled_In** Room (M:1)
- Section **Has** Exam (1:1)

ER Diagram

Course

CourseNo (PK)

Name

Department

|

| Has (1:M)

|

Section (Weak)

SectionNo

Enrollment

|

| Scheduled_In

|

Room

RoomNo (PK)

Capacity

Building

Section

|

| Has

|

Exam

Time

3. University Registrar Office Management System

Entities

Course

- CourseNo (PK)
- Title
- Credits
- Syllabus

Instructor

- InstructorID (PK)
- Name
- Department
- Title

Student

- StudentID (PK)
- Name
- Program

Course_Offering

- Year
- Semester
- SectionNo
- Timings
- Classroom

Relationships

- Course → Course Offering (1:M)
- Instructor → Teaches → Course Offering (M:N)
- Student → Enrolls → Course Offering (M:N)

- Enrollment contains **Grade**
- Course → Prerequisite → Course (Recursive)

ER Diagram

Course

CourseNo

Title

Credits

Syllabus

|

| Offers

|

Course_Offering

Year

Semester

Section

Timing

Classroom

Student ----- Enrolls(Grade) ----- Course_Offering

Instructor ----- Teaches ----- Course_Offering

Course ----- Prerequisite ----- Course

4. Hospital Management System

Entities

Patient

- SS# (PK)
- Name
- Insurance

Physician

- PhysicianID (PK)
- Name
- Specialization

Test_Log

- TestName
- Date
- Time

Relationships

- Doctor Treats Patient (M:N)
- Patient Has Test_Log (1:M)

Tables

Patient

SS#	Name	Insurance

Physician

PhysicianID	Name	Specialization

Test_Log

SS# (FK)	TestName	Date	Time

Doctor_Patient

| PhysicianID (FK) | SS# (FK) |

Patient_History

| SS# | TestName | Date |

ER Diagram

Physician

PhysicianID

Name

Specialization

|

| Treats

|

Patient

SS#

Name

Insurance

|

| Has

|

Test_Log

TestName

Date

Time

5. Company Project Management System

Entities

Department

- DeptID (PK)
- Name

Employee

- SSN (PK)
- Name
- Address
- Salary

- Sex
- BirthDate

Project

- ProjectNo (PK)
- Name
- Location

Dependent (Weak Entity)

- DependentName
- BirthDate
- Relationship

Relationships

- Department Employs Employee (1:M)
- Department Controls Project (1:M)
- Employee Works_On Project (M:N)
 - Attribute: HoursPerWeek
- Employee Supervises Employee (1:M)
- Employee Manages Department (1:1)
 - Attribute: StartDate
- Employee Has Dependent (1:M)

ER Diagram

Department

DeptID

Name

|

| Controls

|

Project

ProjectNo

Name

Location

Department

|

| Employs

|

Employee

SSN

Name

Address

Salary

Sex

BirthDate

|

| Works_On (Hours)

|

Project

Employee ---- Supervises ---- Employee

Employee ---- Has ---- Dependent

6. Schema and Instance

Schema

A schema is the **overall structure/design** of a database.

Example:

Student(StudentID, Name, Age)

This defines the table structure.

Instance

An instance is the **actual data** stored in the database at a particular time.

Example:

StudentID	Name	Age
101	Rahul	20
102	Anita	21

Difference

Logical Schema	Physical Schema	View Schema
Describes database logically	Describes storage details	Shows only required data
Defines tables and relationships	Defines indexes, files, storage	User-specific view
Used by developers	Used by DBA	Used by end users

7. Bank Management System

Entities

Customer

- CustomerID
- Name
- Address
- Phone

Account

- AccountNo
- Type
- Balance

Loan

- LoanNo
- Amount

Branch

- BranchID
- Name
- Location

Relationships

- Customer Owns Account (M:N)
- Branch Maintains Account (1:M)
- Customer Takes Loan (M:N)
- Branch Issues Loan (1:M)

ER Diagram

Customer

CustomerID

Name

Address

Phone

|

| Owns

|

Account

AccountNo

Type

Balance

|

| Maintained By

|

Branch

BranchID

Name

Location

Customer ---- Takes ---- Loan

8. Extended ER (EER) Model

The **Extended Entity-Relationship (EER) Model** is an enhanced version of the ER model that supports inheritance and more complex relationships.

Generalization

Combines similar lower-level entities into a higher-level entity.

Example:

Car

Bus

Truck

↓

Vehicle

Specialization

Divides a higher-level entity into specialized sub-entities.

Example:

Employee

↓

Manager

Engineer

Clerk

Aggregation

Treats a relationship as a higher-level entity to participate in another relationship.

Example:

Employee ---- Works_On ---- Project

↓

Managed_By

↓

Department

9. EER Diagram for Hospital Management System

Superclass

Person

- PersonID
- Name
- Address

Subclasses

Patient

- InsuranceNo

Doctor

- Specialization

Weak Entity

Medical_Record

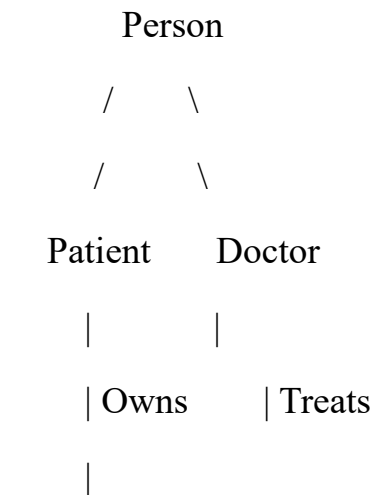
- RecordNo (Partial Key)
- Diagnosis
- Date

Relationships

- Doctor Treats Patient

- Patient Owns Medical_Record

Diagram



Medical_Record (Weak)

RecordNo

Diagnosis

Date

10. Role of DBA (Database Administrator)

A **Database Administrator (DBA)** is responsible for managing, maintaining, securing, and ensuring the efficient operation of databases.

Responsibilities

- 1. Database Installation and Configuration**
 - Install and configure DBMS software.
- 2. Database Design**
 - Create database schemas, tables, indexes, and relationships.
- 3. Security Management**
 - Manage user accounts, roles, and permissions.
 - Protect data from unauthorized access.
- 4. Backup and Recovery**
 - Perform regular backups.
 - Restore data after failures or disasters.
- 5. Performance Monitoring and Tuning**
 - Optimize queries and indexes.
 - Improve database performance.
- 6. Data Integrity**
 - Ensure data accuracy and consistency using constraints.
- 7. Concurrency Control**

- Manage simultaneous access by multiple users.

8. **Storage Management**

- Allocate and monitor disk space.

9. **Troubleshooting**

- Resolve database errors and failures.

10. **Database Maintenance**

- Apply updates, patches, and upgrades.
- Monitor database health and availability.